



Ayan Ali

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● ABOUT ME

Research-focused Electrical Engineer and Master's candidate in Robotics and Artificial Intelligence with a strong foundation in control systems, autonomous navigation, and robotic system integration. Experienced in designing, simulating, and evaluating motion-planning and control pipelines for UGVs and manipulators, including A*, Dynamic Window Approach (DWA), and Model Predictive Control (MPC), primarily in ROS-based environments and MuJoCo simulations.

Hands-on experience in bridging hardware and algorithms, with practical exposure to sensors, actuators, and low-level control, and a systems-engineering approach to building reliable, reproducible robotic workflows from perception and planning to control and performance evaluation. Proficient in Python and MATLAB, with working knowledge of C/C++, control modeling, and simulation-driven development.

Current academic and research interests focus on robust autonomy, adaptive and model-based control, and robotic inspection systems for complex and unstructured environments, with an emphasis on translating theoretical algorithms into deployable, real-world robotic systems.

● PROJECTS

14/08/2025 – 30/01/2026

Object Detection for Autonomous Vehicle Navigation using YOLO

1. Designed and implemented a complete object detection pipeline using YOLOv8, YOLO11, and YOLO12 on the KITTI autonomous driving dataset.
2. Conducted systematic comparative evaluation across architectures using mAP@0.5, mAP@0.5:0.95, precision-recall, FPS, FLOPs, and model size.
3. Addressed GPU memory constraints through resolution scaling, batch size optimization, mixed-precision training, and continued training strategies.
4. Performed class-wise performance analysis for vehicles, pedestrians, and cyclists, identifying challenges in small and rare object detection.
5. Developed perception modules intended for integration with downstream path planning and control systems in autonomous navigation.

07/01/2025 – 06/07/2025

Research on Improving NeRF Training Quality Using Depth Data

I conducted a research project focused on improving NeRF training quality by integrating depth information into the learning pipeline.

The work systematically compared multiple depth-integration strategies including hard geometric constraints, soft depth loss supervision, and hybrid approaches while keeping all other training variables fixed. The goal was to evaluate how different uses of depth data affect rendering fidelity, geometric consistency, and training stability.

The project involved:

1. Designing controlled experiments using real and synthetic depth data
2. Implementing custom training pipelines in Python/PyTorch
3. Quantitative evaluation using PSNR, SSIM, and depth error metrics
4. Analyzing trade-offs between visual quality, geometric accuracy, and computational cost

This research highlights the role of depth as a powerful supervisory signal for 3D perception and scene understanding, with direct relevance to robotic perception, SLAM, and autonomous systems operating in real-world environments.

Link <https://github.com/ayanali827/depth-supervised-nerf-notebooks>

02/02/2025 – 07/07/2025

TriNav – A Tri-Layer Navigation System for UGVs

I developed a simulation-based navigation system for Unmanned Ground Vehicles (UGVs), integrating three powerful layers: global planning, local planning, and trajectory optimization — all tested in the MuJoCo physics simulator.

Key Highlights:

1. A* for global path planning
2. DWA (Dynamic Window Approach) for real-time local obstacle avoidance
3. MPC (Model Predictive Control) for smoother, optimized trajectories
4. PID control for precise low-level velocity tracking
5. DeepSORT for tracking dynamic objects (planned)

Simulation Environment:

Built and tested in MuJoCo with realistic physics and control feedback.

Main Python Libraries:

numpy, scipy, mujoco, ultralytics, deep_sort_realtime, open3d

Link <https://github.com/ayanali827/TriNav-Tri-Layer-Navigation-A-DWA-MPC->

04/10/2022 – 04/07/2023

MODELLING OF A HYBRID PV SYSTEM FOR GRID STABILIZATION

In the realm of renewable energy, our project, "Modeling of a Hybrid PV System for Grid Stabilization," tackles a crucial challenge. It deals with the interaction between photovoltaic (PV) solar systems and grid stability, specifically addressing the issue of Constant Power Loads (CPLs).

CPLs introduce a complex problem due to Incremental Negative Resistance, impacting power quality and grid stability. This challenge is significant across industries, from residential energy efficiency to critical infrastructure support. To harness clean energy effectively, we must find solutions to manage CPLs.

Our project aims to mitigate CPL impact by designing a hybrid PV system. This system includes advanced components like Single-Phase Hybrid Inverters, Phase-Locked Loop (PLL) circuitry, Maximum Power Point Tracking (MPPT) algorithms, and CPL systems. Our methodology involves theoretical study, simulation using Simulink MATLAB, and the creation of precise circuitry models.

Ultimately, this project represents a step towards a more stable and efficient energy future, where solar power and grid stability coexist harmoniously, driving global sustainability efforts.

Link <https://www.linkedin.com/in/ayan-saleem/details/projects/1635541748457/single-media-viewer?>

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● **EDUCATION AND TRAINING**

01/09/2024 – 30/08/2026 Saint Petersburg, Russia

MSC IN ROBOTICS AND ARTIFICIAL INTELLIGENCE ITMO University

Website <https://itmo.ru/> | **Level in EQF** EQF level 7

15/07/2019 – 28/10/2023 Karachi, Pakistan

BACHELOR'S OF ELECTRICAL ENGINEERING NED University Of Engineering and Technology

Website <https://www.neduet.edu.pk/> | **Level in EQF** EQF level 6

01/09/2023 – 28/09/2023

SPECIALIZATION COURSE GOOGLE ADVANCED DATA ANALYTICS Google

Website <https://coursera.org/verify/professional-cert/SCZQFEE3328D>

Website <https://coursera.org/verify/professional-cert/H2BEC848S37Q>

Website https://www.credly.com/badges/e10407a4-82ca-48b6-9255-e3b39d3aa71f/linked_in?t=s4f34m

● **WORK EXPERIENCE**

SAPPHIRE RENEWABLES (WIND POWER COMPANY LTD.) – OPERATIONS MANAGEMENT ENGINEER – 01/02/2024 – 30/09/2024 – KARACHI, PAKISTAN

- 1. Assessing wind turbine data for predictive maintenance.
- 2. Scheduling maintenance to have minimum losses.
- 3. Analyzing wind turbine performance based on site parameters.
- 4. Maintaining data archive for wind turbine statistics.
- 5. Produce detailed documents for all major breakdowns.
- 6. Making detailed daily, monthly and annual reports.
- 7. Corresponding with energy purchaser as per contract.
- 8. Planning and supervising maintenance activities at site.
- 9. Preparing SOPs for plant operations and maintenance procedures.
- 10. Researching and analyzing methods to enhance energy output.

TRAINEE ENGINEER – WINSTON MARRIOT – 20/12/2023 – Current – BUCKINGHAMSHIRE, UNITED KINGDOM

Research and Development Engineer

MAINTENANCE ENGINEERING INTERN – SAPT: SOUTH ASIA PAKISTAN TERMINA – 15/09/2022 – 15/10/2022 – KARACHI, PAKISTAN

- 1. Worked on the electrical maintenance and operation of port infrastructure, supporting ship-to-shore and yard operations.
- 2. Assisted in installation, inspection, and troubleshooting of electrical systems related to port activity monitoring, including CCTV and surveillance camera networks for operational safety and security.
- 3. Participated in preventive and predictive maintenance activities for electrical equipment, contributing to a measurable reduction in unexpected breakdowns.

ENGINEERING INTERN – UNICOL LIMITED – 22/02/2022 – 12/03/2022 – MIRPUR KHAS, PAKISTAN

- 1. Worked on industrial automation systems, focusing on PLC-based control logic and SCADA monitoring for sugar plant operations
- 2. Gained hands-on exposure to SCADA interfaces, including system monitoring, alarm handling, and basic process visualization.
- 3. Supported the operation and maintenance of electrical substations, observing transformers, switchgear, protection relays, and power distribution systems.

● **LANGUAGE SKILLS**

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C2	C1	C1	C1
RUSSIAN	A2	A2	A2	A1	A1

● SKILLS

ROS | Mujoco | Gazebo | Motion Planning | MATLAB | ETAP | Python (Proficient) | C++ | C | Power System Analysis, | Renewable Energy Systems | Python

● HONOURS AND AWARDS

18/12/2024

Obninsk Winter School 2025 – Rosatom (Obninsk, Russia) – MEPHIUniversity

1. Participated in a prestigious international nuclear science program organized by Rosatom, MEPHI, and ObninskTech, collaborating with students and professionals from 39 countries to foster innovation and knowledge exchange.
2. Led Group 3.1 (The Co-Creators), investigating hydrophobicity of low-friction surfaces in steam turbines.
3. Developed a Python GUI to visualize droplet behavior under steam pressure & centrifugal forces.
4. Awarded 2nd Place in the final competition among international teams.

18/03/2025

Obninsk Summer School 2025 – Rosatom (Obninsk, Russia) – MEPHI

1. Engaged in seminars on AI, nuclear, and digital technologies, fostering innovation and collaboration.
2. Built international networks with peers and industry experts.
3. Winner of the 2nd Round of HackAtom 2025, presenting “NukeNet
4. A Gamified Knowledge & Networking Hub” to connect flagship universities.

28/09/2025

World Atomic Week – ROSATOM (Moscow, Russia)

1. Awarded Participation Certificate for the Global HackAtom international student championship during World Atomic Week, contributing to the development of innovative nuclear-technology solutions in a high-intensity, collaborative environment.
2. Represented ITMO university in a global forum of 35+ countries and 1,200+ participants, enhancing cross-cultural teamwork, problem-solving, and technical innovation skills.
3. Engaged with industry experts and advanced nuclear technologies, gaining practical insights into energy systems, R&D approaches, and global innovation trends